Prepared for submission to JHEP nutau-prop / v0.86 / 31 October 2026

A decay–scattering normalization audit of resolved neutrino transition operators

Subhendu Rakshit

Abstract: We resolve a normalization mismatch in the project's implementations of two PKR-like resolved neutrino transition vertices. For a specified massless, distinct-state chiral transition, exact traces establish a factor four between tensor and derivative scattering rates, matching their decay-width ratio. The inherited derivative and tensor DIS kernels were respectively too large by factors sixteen and eight relative to the adopted widths. Fresh CT18 folds and conditional collider screens replace the affected numerical claims. This is a technical correction to the project implementation, not a finding that the published PKR calculation is incorrect or a general exclusion theorem.

Contents

1 Scope and coefficient conventions 1

2 Analytic scattering and crossing checks 1

3 Numerical checks and corrected benchmarks 2

4 Conditional collider screen 3

5 Propagation and interpretation 4

1 Scope and coefficient conventions

The resolved interactions considered in ref. [1] motivate the following explicit vertices for one ordered, distinct-state chiral transition:

$$V_{44}^{\mu} = a(k+k')^{\mu} P_L, \qquad V_{43}^{\mu} = 2aq_{\nu}\sigma^{\nu\mu} P_L, \qquad \kappa = g_q a. \tag{1.1}$$

Here $q = k - k'$, $P_L = (1-\gamma_5)/2$ and $\sigma^{\mu\nu} = i[\gamma^{\mu}, \gamma^{\nu}]/2$, with metric $(+---)$. The tensor's factor two follows from the field strength. We include the Hermitian conjugate, with two distinct conjugate invisible decay channels. The incoming active neutrino has one helicity; only the incoming quark spin is averaged. Singlet-current colour factors cancel.

The project realization is an active-to-sterile transition. Identifying it with a literal charge-conjugated active-flavour operator requires additional field, flavour and multiplicity information. Neither a sterile interpretation nor the original UFO conventions follow from the equation number alone.

An additional exact selection rule matters. For anticommuting fields, $(CP_L)^T = -CP_L$ and $(C\sigma^{\mu\nu}P_L)^T = +C\sigma^{\mu\nu}P_L$ imply that both the charge-conjugated tensor bilinear and the scalar bilinear with an $\overleftrightarrow{\partial}$ derivative are antisymmetric in flavour. They vanish for literally identical diagonal fields. A nonzero derivative realization also needs distinct fields or an off-diagonal assignment; suppressed printed indices do not specify the original numerical model.

2 Analytic scattering and crossing checks

For $\nu(k)q(p) \to N(k')q(p')$, write $s = \hat{s}$, $t = (k-k')^2$, $u = -s - t$ and $K = k + k'$. Up to a phase,

$$\mathcal{M}_{44} = \frac{\kappa}{t - M^2}[\bar{u}(k')P_L u(k)]K_{\mu}[\bar{u}(p')\gamma^{\mu}u(p)]. \tag{2.1}$$

Current conservation and $q \cdot K = 0$ eliminate the longitudinal propagator. The traces give

$$S = \mathrm{Tr}[\not{k}' P_L \not{k} P_R] = -t, \tag{2.2}$$

$$H^{\mu\nu} = \tfrac{1}{2}\,\mathrm{Tr}[\not{p}'\gamma^\mu \not{p}\gamma^\nu], \qquad K_\mu K_\nu H^{\mu\nu} = -4su, \tag{2.3}$$

$$\overline{|\mathcal{M}_{44}|^2} = \frac{4\kappa^2 stu}{(M^2 - t)^2}. \tag{2.4}$$

Using $t = -sy$ and $u = -s(1-y)$,

$$\frac{\mathrm{d}\hat{\sigma}_{44}}{\mathrm{d}y} = \frac{\kappa^2 s^2 y(1-y)}{4\pi(M^2 + sy)^2}. \tag{2.5}$$

The massless external-fermion equations relate the tensor current to twice the derivative current up to a phase. This relation holds for virtual t-channel exchange: it does not assume an on-shell mediator. Therefore

$$\mathrm{d}\sigma_{43} = 4\mathrm{d}\sigma_{44} \quad \text{at equal } a, g_q, M \text{ and phase-space domain.} \tag{2.6}$$

For $s_N = 2m_N E$, $Q^2 = x s_N y$ and $F = \sum_q (x f_q + x f_{\bar{q}})$,

$$\frac{\mathrm{d}^2\sigma_{44}}{\mathrm{d}x\mathrm{d}y} = \frac{\kappa^2 x s_N^2 y(1-y) F(x, Q^2)}{4\pi(M^2 + Q^2)^2}. \tag{2.7}$$

The contact limit has $\hat{\sigma}_{44} = \kappa^2 s^2/(24\pi M^4)$, with relative corrections of order s/M^2.

The vector decay polarization average is $1/3$. The one-channel squared amplitudes are $a^2M^4/3$ and $4a^2M^4/3$. Including both conjugate channels,

$$\Gamma_{44} = \frac{a^2 M^3}{24\pi}, \qquad \Gamma_{43} = \frac{a^2 M^3}{6\pi}. \tag{2.8}$$

These agree with the project's physical widths. In particular,

$$\frac{1}{\Gamma_{\mathrm{inv}}}\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\mathrm{d}y} = \frac{6g_q^2}{M^3}\frac{x s_N^2 y(1-y) F}{(M^2 + Q^2)^2} \tag{2.9}$$

is identical for both vertices at this multiplicity. A coefficient relabelling cannot repair a mismatch in this invariant. Extra invisible species require an explicit change in the width-to-incident-species relation.

The inherited derivative DIS coefficient was $4/\pi$ instead of $1/(4\pi)$. The inherited tensor kernel was twice that value, while its width was four times the derivative width. Relative to the specified widths these kernels overstated the rates by sixteen and eight. The previous width calculation survives; the previous cross-section identification does not.

3 Numerical checks and corrected benchmarks

We used CT18NLO member zero with a leading-order hard process, an isoscalar target, five quark flavours, $Q \geq 1.3$ GeV and $x \geq 10^{-9}$. The integration starts at the physical

Benchmark	g_q	a (GeV^{-1})	$R(100\,\mathrm{PeV})$	$R(220\,\mathrm{PeV})$
v0.73, Eq. 4.4	.030	.00356307	0.3106%	0.3046%
Early Eq. 4.3	.030	.00197258	0.6217%	0.6204%
v0.81, Eq. 4.4	.038	.00237978	0.3179%	0.3155%
v0.83, Eq. 4.3	.017	.00674371	0.6212%	0.5972%

Table 1. Fresh fixed-coupling results at $M = 50$ GeV and $\sqrt{\hat{s}} < 1/a$. The first two 100-PeV NP cross sections are 2.17770×10^{-35} and 4.35948×10^{-35} cm^2.

$Q_{\min}$-dependent y boundary. The reported ratio is the mean of $\sigma_{\mathrm{NP}}/\sigma^{\nu}_{\mathrm{SM}}$ and $\sigma_{\mathrm{NP}}/\sigma^{\bar{\nu}}_{\mathrm{SM}}$, with SM including CC and NC. It is not a ratio to NC alone or a detector event ratio.

Changing the cutoff to $0.5/a$ at fixed couplings lowers the first two 100-PeV ratios to 0.1232% and 0.3416%. Since $a = c/\Lambda$, the choice $1/a$ is a convention for the Wilson coefficient, not a universally established UV threshold. Equal amplitudes require $a_{44} = 2a_{43}$. At a common physical cutoff both rates and widths then agree; imposing $1/a$ separately gives different domains and a prescription-dependent hierarchy. This is not a proof of equivalence between complete electroweak or UV theories.

Exact symbolic matrix traces pass at general scattering angle and arbitrary decay mass, including the Ward identities. Twenty tests check the contact limit, endpoints, coefficient mapping and analytic coupling bounds. Fourteen Mac Studio worker processes, each with single-thread numerical libraries, computed 288 base folds and targeted refinements. The largest relative quadrature change at refined maxima was 1.42×10^{-7}, and the largest interpolation error was 1.93×10^{-4}. These figures do not include PDF, perturbative, low-Q or experimental uncertainties.

4 Conditional collider screen

At fixed mass and in the narrow-width approximation, let $z = g_q^2$, $I = \Gamma_{\mathrm{inv}}$, $B = \Gamma_{qq}/g_q^2$ and

$$P = \frac{zI}{Bz + I}, \qquad r = K_M P, \qquad D = z - P = g_q^2 \mathrm{BR}_{qq}. \tag{4.1}$$

The bounds on z can be solved analytically. A monojet condition $P \le P_{\max}$ gives $z \le P_{\max} I/(I - BP_{\max})$ only if $I > BP_{\max}$. A dijet condition $D \le d$ gives

$$z \le \tfrac{1}{2}\left[d + \sqrt{d^2 + 4dI/B}\right]. \tag{4.2}$$

We retain $\Gamma/M \le 0.1$, $g_q^2 \le 4\pi$ and $1/a > M$. Every sampled a interval was optimized and its winning value checked by a direct fold, avoiding a first-crossing-only scan.

CMS reports universal-vector dijet coupling limits from 0.03 to 0.13 over 50–300 GeV, assuming a fully hadronic decay [2]. We deliberately use the weakest value, $g_q\sqrt{\mathrm{BR}_{qq}} \le 0.13$, at each tested mass. This relaxes the mass-dependent constraint under the narrow-width mapping. The resulting nominal $r \le 1$ maxima are:

M (GeV)	Eq. 4.4	Eq. 4.3
50	3.142%	4.514%
100	0.2197%	0.2635%
220	0.0583%	0.0743%

These are conditional screens, not exact 95% CL exclusions. The $M = 50$ GeV normalization uses the existing 100k-event same-seed reference; 100 and 220 GeV use 20k-event pilot shapes. Exact mass-specific dijet data were not recovered. With the same deliberately weak dijet envelope but $r \leq 1.2$, Eq. 4.3 at 50 GeV reaches 5.258%. Thus the relaxed nominal bound must not be promoted to a robust exclusion. Approximate mass-specific inputs yield smaller effects, but remain sensitivity diagnostics. Other mediator masses are not closed.

At the same couplings, correcting DIS leaves the existing collider r unchanged because the widths already had this normalization. What changes is whether that point satisfies the UHE target. The $r = 1.239$ benchmark is no longer a 5% UHE benchmark.

5 Propagation and interpretation

For the specified sterile final state there is no SM interference and the new transport term is an active-flux sink. The unregenerated optical depth $N_A X \sigma_{\rm NP}$ inherits the rate correction. An already regenerated SM spectrum cannot simply be multiplied by a sink exponential: NC downscattering and tau decay couple energies. We do not claim a new transport solution or a tau-specific IceCube likelihood here.

The corrected benchmark effects remain subpercent at 220 PeV. A large relative change on an opaque vertical chord is not, by itself, an observable cosmogenic event enhancement. The original objective remains a valid model with measurable tau-neutrino propagation distortions after laboratory, IceCube, validity and detector constraints. These benchmarks do not establish that goal, but neither do three mass screens prove a general no-go theorem.

The error established here belongs to the project's normalization chain. An attribution to the published PKR analysis requires reproducing its actual field conventions and collider model. Modern experimental constraints and implementation errors must be discussed separately. All superseded results are preserved, with corrected code and machine-readable outputs in v0.86.

The concurrently added published-bound reconstruction. PKR quotes 5-GeV IceCube bounds $\kappa_{43} = 4.8 \times 10^{-4}$ and $\kappa_{44} = 2.5 \times 10^{-4}$ GeV^{-1} [1]. If the two bounds use an identical allowance, flavour sum and the coefficient convention above, they imply $F_{43}/F_{44} = (2.5/4.8)^2 = 0.271267$ for $\sigma_i = \kappa_i^2 F_i$, instead of four. Our kernels evaluated at those couplings have ratio 14.7456. This is a conditional normalization discrepancy, but the pair does not determine which absolute implementation is wrong. Trusting the 4.3 anchor gives a 4.4 bound of 9.6×10^{-4}; trusting the 4.4 anchor gives a 4.3 bound of 1.25×10^{-4} GeV^{-1}. Neither is a new IceCube fit. The concurrent v0.85 note's inferred coefficient 58.98 has rounding range 55.5–62.7, which does not contain 64. Extra uncertainty is necessary to

support that suggested identification. This distinction separates the directly demonstrated repository error from a hypothesis about the published implementation.

References

[1] S. Pandey, S. Karmakar and S. Rakshit, *Strong constraints on non-standard neutrino interactions: LHC vs. IceCube*, arXiv:1907.07700.

[2] CMS Collaboration, *Search for low-mass vector and scalar resonances decaying into a quark-antiquark pair in proton-proton collisions at* $\sqrt{s} = 13$ *TeV*, CMS-EXO-24-007, arXiv:2603.21965; official results and branching-ratio assumptions.